# SUPERSTORE SALES PROFITABILITY ANALYSIS

## Business Analysis Report

**Prepared by: Karthik Siva | April 2026**

**EXECUTIVE SUMMARY:**

Superstore's overall profit margin stands at ~12%, but significant losses are concentrated in the Central region, Furniture category, and high-discount orders. Immediate action on discount policy and product mix can recover an estimated 15–20% of lost profit.

**BUSINESS CONTEXT**

The VP of Sales raised concerns about declining profitability despite steady revenue growth. This analysis examines 9,994 orders across 4 regions and 3 product categories (2014–2017) to identify the root causes of profit leakage.

**Dataset:** Sample Superstore (Kaggle) — 9,994 orders, 21 variables

**Tools:** Python (Pandas), SQL, Power BI

**KEY FINDINGS**

**Finding 1:**

- Central Region is the Weakest Performer
- Central region profit margin: 7.92%
- Company average margin: ~12%
- Central is 34% below the company average despite similar sales volume
- This means Central is generating revenue but not converting it to profit — a classic margin leak problem

**Finding 2:**

- Tables Sub-Category is Loss-Making
- Tables generated a net loss of $17,725 over 4 years
- Every table order on average COSTS the company money
- Bookcases also underperform with a near-zero margin
- Furniture as a whole drags category profitability down

**Finding 3**

- Discounting is Directly Destroying Profit
- Correlation between Discount and Profit: -0.219
- Orders with 0% discount average +$65 profit
- Orders with 40%+ discount average -$90 profit (a loss)
- The company is literally paying customers to take products

**Finding 4**

- Ohio is the Worst Performing State
- Ohio ranks as the highest loss-generating state
- Concentrated losses suggest either pricing, competition, or discount mismanagement at the local sales level

**ROOT CAUSE ANALYSIS**

All 4 findings point to ONE underlying cause:

- Discount decisions are being made without profit guardrails.
- Sales reps are offering high discounts (40%+) to close deals, particularly in the Central region and on Furniture items. This inflates order volume but destroys margin. The business is optimising for revenue, not profit.

**RECOMMENDATIONS:**

**Recommendation 1**: Implement a Discount Cap Policy [HIGH PRIORITY]

- No discount above 30% without regional manager approval
- No discount above 40% without VP approval
- Automate this rule in the CRM/order management system
- Expected outcome: Eliminate loss-making orders (~8% of all orders)

**Recommendation 2**: Review Furniture Category Strategy[MEDIUM PRIORITY]

- Conduct SKU-level profitability review for Tables & Bookcases
- Either renegotiate supplier costs or discontinue loss SKUs
- Redirect sales focus to Office Supplies (highest margin category)
- Expected outcome: Category margin improvement of 3–5%

**Recommendation 3**: Central Region Intervention [HIGH PRIORITY]

- Audit all orders with discount > 30% in Central region
- Replace volume-based sales targets with margin-based targets
- Provide sales reps with real-time margin visibility per order
- Expected outcome: Central margin improves from 7.92% to ~11%

**Recommendation 4**: Ohio Market Review [LOW PRIORITY]

- Investigate whether Ohio losses are pricing or competition-driven
- Consider reducing marketing spend in Ohio until the root cause is found
- Expected outcome: Reduce state-level losses by 40%

**EXPECTED IMPACT**

If Recommendations 1 and 3 are implemented together:

- Estimated profit recovery: $35,000–$50,000 annually
- Loss-making orders eliminated: ~800 orders/year
- Overall margin improvement: 12% → 14–15%

These are conservative estimates based on historical order data. Actual impact depends on sales team compliance and CRM enforcement.